THE SIX-TERM JACOBI-ZARISKI SEQUENCE CLOSES WITH A ZERO WHEN THE LAST MAP IS A LOCAL COMPLETE INTERSECTION

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Allowed standard prerequisites (editorial).

- The six-term exact Jacobi-Zariski sequence of a composition of ring maps. Algebra, Lemma \ref{algebra-lemma-exact-sequence-NL}, cited at proof lines 8488-8490. Provides the sequence that is completed by a zero.
- The conormal sequence for an ideal cut out by an H1-regular sequence, giving the intersection of the extended ideal with the square of the kernel. Lemma \ref{lemma-conormal-sequence-H1-regular-ideal}, cited at proof lines 8497-8499. Identifies the kernel of the comparison map with the extended ideal modulo the product.

Source reference. Original lemma 07D4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9399–9399. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00S2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1024–1024. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07CX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9392–9392. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $A \to B \to C$ be ring maps. Assume $B \to C$ is a local complete intersection homomorphism. Choose a presentation $\alpha : A[x_s, s \in S] \to B$ with kernel I. Choose a presentation $\beta : B[y_1, \ldots, y_m] \to C$ with kernel J. Let $\gamma : A[x_s, y_t] \to C$ be the induced presentation of C with kernel K. Then we get a canonical commutative diagram*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \Omega_{A[x_s]/A} \otimes C & \longrightarrow & \Omega_{A[x_s,y_t]/A} \otimes C & \longrightarrow & \Omega_{B[y_t]/B} \otimes C & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & I/I^2 \otimes C & \longrightarrow & K/K^2 & \longrightarrow & J/J^2 & \longrightarrow & 0 \end{array}$$

with exact rows. In particular, the six term exact sequence of Algebra, Lemma 00S2 can be completed with a zero on the left, i.e., the sequence

$$0 \to H_1(NL_{B/A} \otimes_B C) \to H_1(L_{C/A}) \to H_1(L_{C/B}) \to \Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to \Omega_{C/B} \to 0$$

is exact.

Proof. The only thing to prove is the injectivity of the map $I/I^2 \otimes C \to K/K^2$. By assumption the ideal J is Koszul-regular. Hence we have $IA[x_s, y_j] \cap K^2 = IK$ by Lemma 07CX. This means that the kernel of $K/K^2 \to J/J^2$ is isomorphic to $IA[x_s, y_j]/IK$. Since $I/I^2 \otimes_A C = IA[x_s, y_j]/IK$ by right exactness of tensor product, this provides us with the desired injectivity of $I/I^2 \otimes_A C \to K/K^2$. $\square$